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Sensing the World
Project Reflection

For our final project, my group created a proof-of-concept prototype for a bookcase that could help protect old books and other media from moisture and light damage. The goal was to create a system that could monitor the conditions inside the bookcase and show how sensors could be used to preserve materials over time. We used an Arduino Nano ESP32 as the microcontroller along with a photosensor and a moisture sensor that we had made earlier in the course. Throughout the project, we spent time wiring everything together, testing the sensors, and making sure the code worked correctly. There was a lot of trial and error involved, but over time we got the system working the way we wanted. Once the prototype was complete, we created a slideshow and presented our project to the class.

One thing I learned during this project was how helpful AI can be when coding. Before this project, I did not have much experience using AI for programming, but it made it easier to figure out problems and understand parts of the code that I was stuck on. I also learned a lot more about wiring sensors and connecting them to a microcontroller. It was cool to actually see the sensors respond to changes in the environment instead of just reading about how they work. Being able to build something physical made the concepts from class feel much more real.

This project also gave me more experience working in a group. We had to split up tasks, communicate with each other, and make sure everything came together by the deadline. Whenever we ran into problems, we usually had to troubleshoot them as a team and try a few different solutions before finding one that worked. For example, our original stepper motor pulley system wasn’t working properly so we switched to a servo motor and lever system which worked way better.

Overall, I think the project was successful and worked well as a proof of concept. The final prototype showed that a system like this could potentially be used to protect books and other media from moisture and light exposure. My favorite part was seeing everything finally come together after spending so much time working on individual pieces. Presenting the finished project was rewarding because it showed how much progress we had made from the original idea. This project gave me more confidence working with sensors, coding, and electronics, and it showed me how different parts of engineering come together to solve a real problem.